

Hongyu Cao

School of Computing and Augmented Intelligence, Arizona State University, Tempe, AZ 85281, USA.

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EDUCATION

PhD student of Computer Science Arizona State University,	Januray 2026 -
Master of Computer Science University of Science and Technology of China, GPA: 3.72/4.30, top 30%	August 2022 - May 2025
Bachelor of Computer Science University of Science and Technology of China, GPA: 3.59/4.30, top 30%	August 2018 - May 2022
HSC, Class XII Xi'an Shi Tieyi Senior High, China, Shanxi, top 0.5%	June 2015 - April 2018
SSC, Class X Xi'an Shi Tieyi Junior High, China, Shanxi, top 0.5%	March 2012 - March 2015

SKILLS AND INTERESTS

Skills	Research Methods: Auto Feature Transformation, Latent Diffusion, Guided Generation LLMs & Agents: LoRA Fine-tuning, RAG, Agentic Reasoning Language Proficiency: TOEFL (Speaking: 21, Writing: 23, Listening: 27, Reading: 29)
Interests	Data-Centric AI; Spatio-Temporal Data Mining; Representation Learning; LLM & Agentic AI; Reinforcement Learning

PUBLICATION

Hongyu Cao , David A. King, Xinyuan Wang, Arun Vignesh Malarkkan, Kunpeng Liu, Dongjie Wang, Yanjie Fu, <i>Parking Without Building: Agentic AI for Parking Planning Through Urban Contradiction Reasoning</i> (Preprint) 2026
Hongyu Cao , Kunpeng Liu, Dongjie Wang, Yanjie Fu, <i>Mitigating Shortcut Reasoning in Language Models: A Gradient-Aware Training Approach</i> (arXiv:2603.20899) 2026
Hongyu Cao , Jinghan Zhang, Kunpeng Liu, Dongjie Wang, Feng Xia, Haifeng Chen, Xia Hu, Yanjie Fu, <i>Sim2Act: Robust Simulation-to-Decision Learning via Adversarial Calibration and Group-Relative Perturbation</i> (arXiv:2603.09053) 2026
Hongyu Cao , Haoyue Bai, Yanjie Fu, <i>Structured Memory and Role-Aware Decision Making for Supply Chain Transportation</i> (IEEE BigData 2025) 2025
Hongyu Cao , Jinlong Li, <i>Desentiment: A New Method to Control Sentimental Tendency During Summary Generation</i> (Information 16(6):453) 2025
Weiye Zhang, Jiancheng Yang, Ruoyu Chen, Siyu Huang, Pusheng Xu, Xiaolan Chen, Ziwei Wang, Shanfu Lu, Hongyu Cao , et al., <i>Fundus to Fluorescein Angiography Video Generation as a Retinal Generative Foundation Model</i> (npj Digital Medicine) 2026

INTERNSHIP/TRAININGS

Natural Language Processing Internship , Iflytek ,Institute of Artificial Intelligence, Hefei	October 2022
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ACHIEVEMENTS

USTC Scholarship	Autumn 2022
USTC Scholarship	Spring 2023
USTC Scholarship	Autumn 2023

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.